

Education

Ph.D. in Applied Mathematics

University of Colorado Boulder
3rd-year Ph.D. student, GPA: 4.0

Expected May 2028
Boulder, CO

M.S. in Applied Mathematics

University of Colorado Boulder

May 2026
Boulder, CO

B.S. in Applied Mathematics, *magna cum laude*

University of Colorado Boulder
Concentration in Aerospace Engineering

May 2023
Boulder, CO

Research Interests

- Numerical PDEs
- Multiscale modeling/simulation
- Particle fluid flows
- Numerical ODEs
- Stiff and/or oscillatory systems
- Transient dynamics in power systems
- Functional, asymptotic, and numerical analysis

Research Experience

Department of Applied Mathematics

Research Assistant

University of Colorado Boulder

May 2022–present

MOBILITY PROBLEM (PARTICLE STOKES FLOW)

Project advisor: [Eduardo Corona](#)

August 2025–present

- Investigated boundary integral methods for elliptic PDE to be applied to the *Mobility problem*, which describes how rigid particles move in highly viscous (Stokes) flows.
- Successfully accelerated the existing implementation of the chosen method.
- Developed a new technique that improves the conditioning of the boundary integral operator.

FAST SOLVERS FOR POWER SYSTEMS APPLICATIONS

May 2023–present

Project advisor: [Eduardo Corona](#)

Project collaborators: [Fiona Majeau](#), [Bri-Mathias Hodge](#), [Amir Sajadi](#)

- Learned fundamental concepts of power system dynamics.
- Studied and implemented various solvers for stiff and/or oscillatory systems of ODEs.
- Wrote code to perform electromagnetic transient simulations using systems and machine parameters from [NREL's Sienna package](#).

GENTRIFICATION DYNAMICS

May 2022–September 2024

Project advisors: [Juan Restrepo](#) and [Nancy Rodríguez](#)

- Developed a dynamical system model of gentrification.
- Performed functional and numerical analysis to understand observed dynamics as a function of model parameters.

CONDUIT PDE WAVEPACKET INTERACTIONS

August 2023–September 2023

Advisor: [Mark Hoefer](#)

- Obtained asymptotic descriptions of small amplitude wavepackets interacting with rarefaction and dispersive shock waves in the context of the [conduit PDE](#).

Los Alamos National Laboratory

Intern - Physics XTD-PRI

Los Alamos, NM

May 2018–August 2018

ALGEBRAIC TOPOLOGY FOR COMPUTATIONAL FLUID DYNAMICS

Project advisor: [Roy S. Baty](#)

- Researched fluid dynamics through the scope of algebraic topology.
- Analyzed Schlieren images of supersonic airflow from an elliptic nozzle in order to establish a relationship between the topological characteristics and the time-varying pressure fields of the flow.
- A talk on this research was delivered at the 71st Annual Meeting of the APS Division of Fluid Dynamics (abstract [here](#)).

Publications

Peer-Reviewed Articles

1. **Shaw, J. D.**; Restrepo, J. G.; Rodríguez, N., *A dynamical model of gentrification: Exploring a simple rent control strategy*, Phys. Rev. E, 2024 (<https://arxiv.org/abs/2406.12092>) [3 citations]

Preprints & Other

1. Trujillo, M.; Sajadi, A.; **Shaw, J. D.**; Hodge, B.-M., *Analytical Models of Frequency and Voltage in Large-Scale All-Inverter Power Systems*, Journal TBD 2026 (<https://arxiv.org/pdf/2501.10592>)

[0 citations]

2. Trujillo, M.; Sajadi, A.; **Shaw, J. D.**; Hodge, B.-M., *Computationally Efficient Analytical Models of Frequency and Voltage in Low-Inertia Systems*, IEEE Transactions on Sustainable Energy 2025 (<https://arxiv.org/abs/2506.06620>) [0 citations]

Talks & Presentations

Talks

- March 2025, Complex/Dynamical Systems Seminar, University of Colorado Boulder (invited talk)
- January 2025, Dynamics Days US 2025, Denver, CO (contributed talk, travel award recipient)
- October 2023, Mathematical Biology Seminar, University of Colorado Boulder (contributed talk)

Poster Presentations

- January 2025, Dynamics Days US 2025, Denver, CO (contributed poster)
- January 2023, Dynamics Days US 2023, Denver, CO (contributed poster)

Teaching Experience

Department of Applied Mathematics

Teaching Assistant

University of Colorado Boulder

August 2022–May 2025

APPLIED ANALYSIS (APPM [5440/5450](#)) (GRADUATE COURSES)

August 2024–May 2025

Instructor: [François G. Meyer](#)

VECTOR CALCULUS (APPM [2350](#))

May 2024–May 2025

Instructors: [Robert Benim](#), [Eric Thaler](#)

CALCULUS 1 (APPM [1350](#))

August 2023–May 2024

Instructor: [Robert Benim](#)

DISCRETE MATH (APPM [3170](#))

August 2022–December 2022

Instructor: [François G. Meyer](#)

Department of Mathematics

Supplemental Instructor, vector calculus

John Tyler Community College, Richmond, VA

January 2019–May 2019

Leadership & Service

Society for Industrial and Applied Mathematics (SIAM) University of Colorado Boulder
President, Graduate Student Chapter August 2025–May 2026

- Organized biweekly applied mathematics graduate and alumni talks, national laboratory tours, and graduate student study sessions.
- Led the chapter in financial planning and budgeting.

Industry Liaison, Graduate Student Chapter August 2024–May 2025

- Helped coordinate biweekly applied mathematics graduate and alumni talks.

Buckskin National Youth Leadership Training Richmond, VA
Senior Staff Member January 2016–August 2023

- Volunteered with other senior staff members throughout the year to optimize the curriculum and execution of an annual seven-day leadership seminar.
- Gave talks on leadership methodologies and personal leadership development.

Skills

Software: MATLAB, , C++,  python[™], L^AT_EX
Interpersonal: Public speaking, leadership, conflict resolution
Hobbies: Trail running, rock climbing, skiing, reading, cooking

References

Eduardo Corona <i>Advisor</i>	University of Colorado Boulder <i>Department of Applied Mathematics</i>
Bri-Mathias Hodge <i>Collaborator</i>	University of Colorado Boulder <i>Department of Electrical, Computer and Energy Engineering</i>
Juan Restrepo <i>Collaborator</i>	University of Colorado Boulder <i>Department of Applied Mathematics</i>
Nancy Rodríguez <i>Collaborator</i>	University of Colorado Boulder <i>Department of Applied Mathematics</i>
François G. Meyer <i>Teaching reference</i>	University of Colorado Boulder <i>Department of Applied Mathematics</i>